EXPLORATORY DATA ANALYSIS AND DEEP LEARNING FOR BANK MARKETING

Case Study: Predicting Term Deposit Subscription

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Abstract

This project explores the UCI Bank Marketing dataset to understand customer characteristics and patterns associated with term-deposit subscription. The dataset contains client information and details about direct marketing contacts. The workflow includes data inspection, duplicate checking, treatment of unknown categorical values, numerical scaling, categorical encoding, exploratory visualizations, and training a MultiLayer Perceptron (MLP) classifier. The dataset is divided into training, validation, and test sets. Model performance is assessed using accuracy, a classification report, a confusion matrix, and ROC-AUC. Five exploratory visualizations are used to study target balance, age distribution, balance outliers, subscription across job types, and numerical correlations. After running the supplied code, insert the actual model results and observations in the result sections. The goal is to demonstrate a reproducible data-analysis workflow and identify useful patterns without claiming unsupported results.

1. Introduction and Objectives

Direct marketing allows banks to contact customers about financial products. Understanding which customer groups are more likely to subscribe to a term deposit can help a bank study campaign outcomes and plan future outreach.
Objectives:

- Understand the structure and quality of a real-world dataset.
- Clean and preprocess numerical and categorical variables.
- Use at least five visualizations to explore distributions and relationships.
- Train a Deep Learning MLP classifier.
- Evaluate the model with multiple metrics and graphs.
- Record limitations and possible future improvements.

2. Dataset Description

Dataset name: Bank Marketing.
Source: UCI Machine Learning Repository, https://archive.ics.uci.edu/dataset/222/bank+marketing
Application: Predict whether a contacted client subscribed to a term deposit.
Dataset file: bank-full.csv. The official UCI description reports 45,211 instances and 17 input features for this version. The target column is `y`, with values `yes` and `no`.
Examples of input fields include age, job, marital status, education, default status, balance, housing loan, personal loan, contact type, day, month, call duration, campaign, pdays, previous contacts, and previous campaign outcome.
Important limitation: call duration is only known after a call. For a real pre-call prediction system, it should be excluded to avoid using information unavailable at prediction time. This starter code keeps it for a coursework benchmark; mention this limitation in the report.

3. EDA and Preprocessing

The code displays the first rows, column names, data types, descriptive statistics, missing-value counts, and duplicate count. Duplicate rows are removed. The category value `unknown` is treated as missing, then numerical missing values are imputed with the median and categorical missing values with the most frequent category.
Numerical columns are standardized using StandardScaler. Categorical columns are converted into indicator columns using one-hot encoding. The dataset is split into 70% training, 15% validation, and 15% test data with stratification so the target proportions are approximately preserved. Preprocessing is fitted only on the training set to reduce data leakage.

4. Data Visualization and Interpretation

Figure 1 — Target distribution: Shows the number of clients who did and did not subscribe. Use it to discuss whether the target classes are balanced.
Figure 2 — Age distribution: Shows the distribution of customer ages. Describe the age ranges with more observations and whether the distribution is skewed.
Figure 3 — Balance boxplot: Helps identify spread and possible outliers in yearly balance. Outliers are not automatically errors and should be interpreted carefully.
Figure 4 — Subscription by job: Compares subscription counts across job categories. Compare the counts carefully; categories with more customers may have more subscriptions simply because they are larger.
Figure 5 — Numerical correlation heatmap: Displays pairwise linear correlations among numerical variables. Correlation does not prove causation.
After running the script, paste each generated PNG from the `outputs` folder into this section and add 2–3 sentences describing the actual pattern visible in your graph.

5. Deep Learning Model

Model: Multi-Layer Perceptron (MLP) classifier.
Task: Binary classification — `no` (0) or `yes` (1).
Architecture: The script compares two hidden-layer configurations: (32,) and (64, 32), using ReLU activation and the Adam optimizer.

Training settings: 15 training passes (epochs), batch size 256, learning rate 0.001, and L2 regularization parameter alpha = 0.0001. The configuration with the higher final validation accuracy is selected.
Hyperparameter tuning is basic and limited to the hidden-layer configurations. The script also estimates encoded-feature importance by measuring the decrease in accuracy after shuffling each feature on a sample of the test data.

6. Model Results and Evaluation

Insert the actual values from `outputs/model_metrics.txt` after running the code.
Selected hidden layers: [Fill from output]
Validation accuracy: [Fill from output]
Test accuracy: [Fill from output]
ROC-AUC: [Fill from output]
Figure 6 — Training loss vs epoch: Shows how the training loss changes over training passes. A decreasing curve generally indicates that the model is fitting the training data.
Figure 7 — Validation accuracy vs epoch: Shows validation performance during training. Large fluctuations or a falling trend may indicate unstable learning or overfitting.
Figure 8 — Confusion matrix: Shows correct and incorrect predictions for each class. Discuss false positives and false negatives, not only total accuracy.
Figure 9 — ROC curve: Shows the trade-off between true-positive rate and false-positive rate across thresholds. AUC closer to 1 indicates stronger ranking performance; 0.5 is similar to random ranking.
Figure 10 — Feature importance: Shows the top encoded features whose shuffling most reduces accuracy. This is a model-specific diagnostic, not proof that a feature causes subscription.

7. Conclusion and Future Scope

This case study provides a complete workflow for exploring the Bank Marketing dataset and building an MLP classifier. The final conclusion must be completed after inspecting the generated graphs and actual metrics.
Key findings from the run: [Write 2–3 observations based on the actual graphs.]
Model performance: [Summarize actual test accuracy, ROC-AUC, and confusion matrix.]
Challenges: Categorical encoding, missing/unknown values, class imbalance, and avoiding data leakage are important considerations.
Future scope: Compare the MLP with logistic regression and tree-based models; use stratified crossvalidation; tune more hyperparameters; evaluate precision, recall, and F1-score; consider class weighting; and exclude call duration when making predictions before a call.

8. References

1. Moro, S., Rita, P., & Cortez, P. (2014). Bank Marketing [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5K306
2. Moro, S., Laureano, R., & Cortez, P. (2011). Using data mining for bank direct marketing: An application of the CRISP-DM methodology. Proceedings of the European Simulation and Modelling Conference.
3. Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

4. McKinney, W. (2010). Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference.
5. Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95.

Appendix A — Code and Repository

Source code: bank_marketing_eda.py
README: README.md
Generated figures and metrics: outputs/
GitHub repository: [Paste your public GitHub repository URL here]
Before submission, run the code, insert real graphs and metrics into this report, verify the results, and export the final report to PDF or Word.

Output and Results:

```
Downloading Bank Marketing dataset...

Dataset loaded successfully!
Dataset shape: (45211, 17)

First five rows:
```

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previous	poutcome	y
0	58	management	married	tertiary	no	2143	yes	no	unknown	5	may	261	1	-1	0	unknown	no
1	44	technician	single	secondary	no	29	yes	no	unknown	5	may	151	1	-1	0	unknown	no
2	33	entrepreneur	married	secondary	no	2	yes	yes	unknown	5	may	76	1	-1	0	unknown	no
3	47	blue-collar	married	unknown	no	1506	yes	no	unknown	5	may	92	1	-1	0	unknown	no
4	33	unknown	single	unknown	no	1	no	no	unknown	5	may	198	1	-1	0	unknown	no

```
Column names:
```

```
['age', 'job', 'marital', 'education', 'default', 'balance', 'housing', 'loan',
'contact', 'day', 'month', 'duration', 'campaign', 'pdays', 'previous',
'poutcome', 'y']

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45211 entries, 0 to 45210
Data columns (total 17 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   age        45211 non-null  int64
 1   job        45211 non-null  object
 2   marital    45211 non-null  object
 3   education  45211 non-null  object
 4   default    45211 non-null  object
 5   balance    45211 non-null  int64
 6   housing    45211 non-null  object
 7   loan       45211 non-null  object
 8   contact    45211 non-null  object
 9   day        45211 non-null  int64
 10  month      45211 non-null  object
 11  duration   45211 non-null  int64
 12  campaign   45211 non-null  int64
 13  pdays      45211 non-null  int64
 14  previous   45211 non-null  int64
 15  poutcome   45211 non-null  object
 16  y          45211 non-null  object
dtypes: int64(7), object(10)
memory usage: 5.9+ MB

Statistical summary:
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
age	45211.0	NaN	NaN	NaN	40.93621	10.618762	18.0	33.0	39.0	48.0	95.0
job	45211	12	blue-collar	9732	NaN	NaN	NaN	NaN	NaN	NaN	NaN
marital	45211	3	married	27214	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education	45211	4	secondary	23202	NaN	NaN	NaN	NaN	NaN	NaN	NaN
default	45211	2	no	44396	NaN	NaN	NaN	NaN	NaN	NaN	NaN
balance	45211.0	NaN	NaN	NaN	1362.272058	3044.765829	-8019.0	72.0	448.0	1428.0	102127.0
housing	45211	2	yes	25130	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
loan	45211	2	no	37967	NaN	NaN	NaN	NaN	NaN	NaN	NaN
contact	45211	3	cellular	29285	NaN	NaN	NaN	NaN	NaN	NaN	NaN
day	45211.0	NaN	NaN	NaN	15.806419	8.322476	1.0	8.0	16.0	21.0	31.0
month	45211	12	may	13766	NaN	NaN	NaN	NaN	NaN	NaN	NaN
duration	45211.0	NaN	NaN	NaN	258.16308	257.527812	0.0	103.0	180.0	319.0	4918.0
campaign	45211.0	NaN	NaN	NaN	2.763841	3.098021	1.0	1.0	2.0	3.0	63.0
pdays	45211.0	NaN	NaN	NaN	40.197828	100.128746	-1.0	-1.0	-1.0	-1.0	871.0
previous	45211.0	NaN	NaN	NaN	0.580323	2.303441	0.0	0.0	0.0	0.0	275.0
poutcome	45211	4	unknown	36959	NaN	NaN	NaN	NaN	NaN	NaN	NaN
y	45211	2	no	39922	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
Missing values:
age          0
job          0
marital      0
education    0
default      0
balance      0
housing      0
loan         0
contact      0
day          0
month        0
duration     0
campaign     0
pdays        0
previous     0
poutcome     0
y            0
dtype: int64

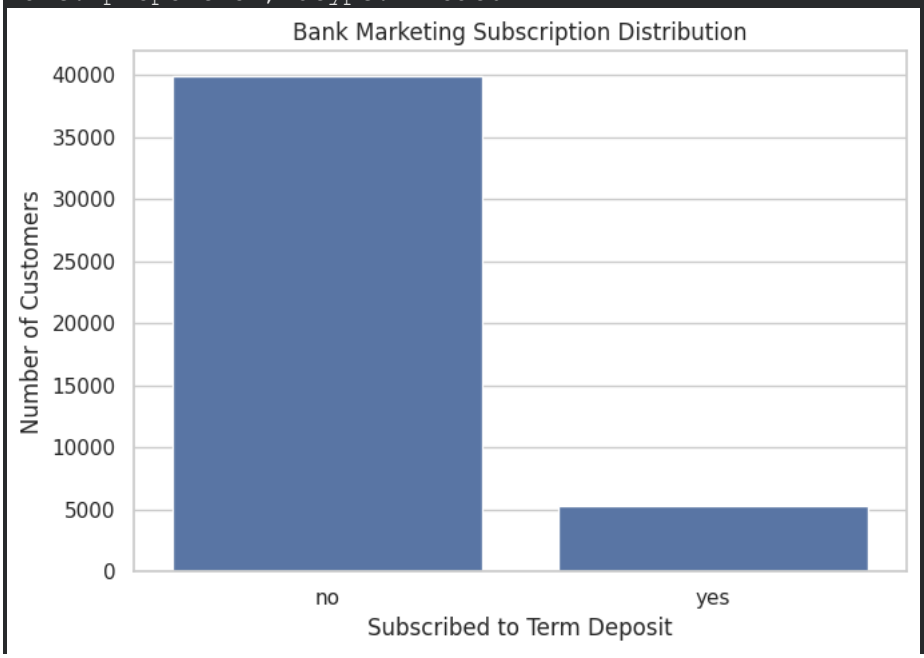
Duplicate rows: 0
```

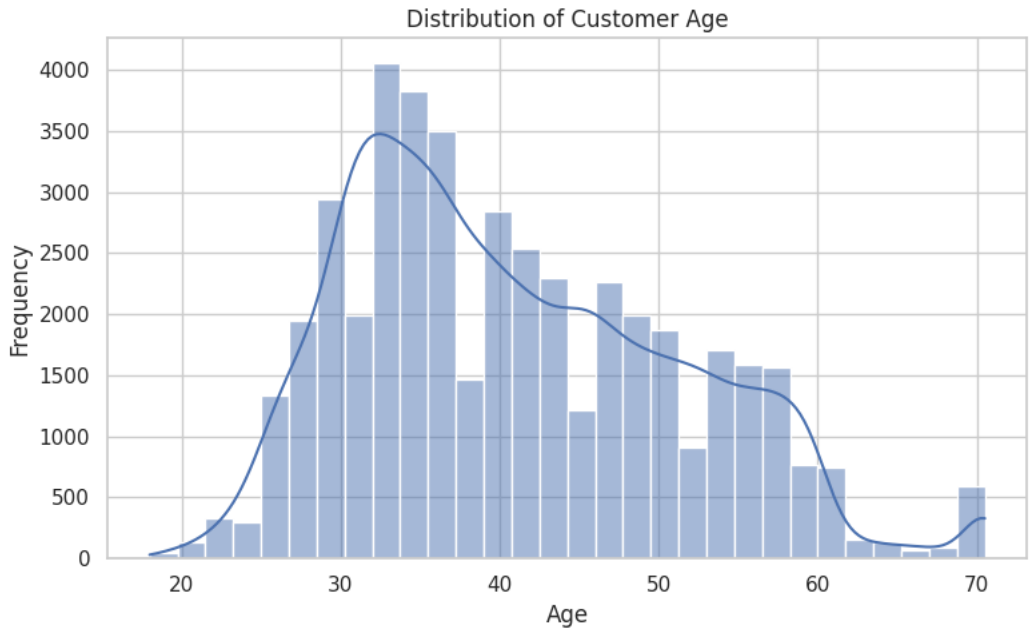
```
Missing values after cleaning:
0

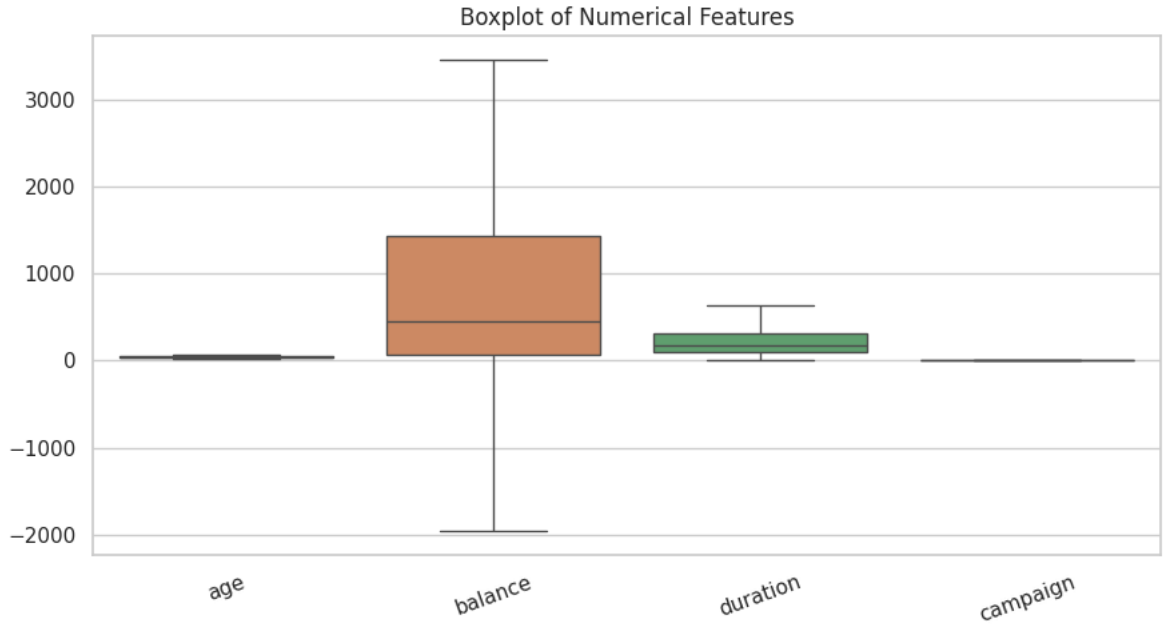
Outlier counts using the IQR method:
age : 487
balance : 4729
day : 0
duration : 3235
campaign : 3064
pdays : 8257
previous : 8257

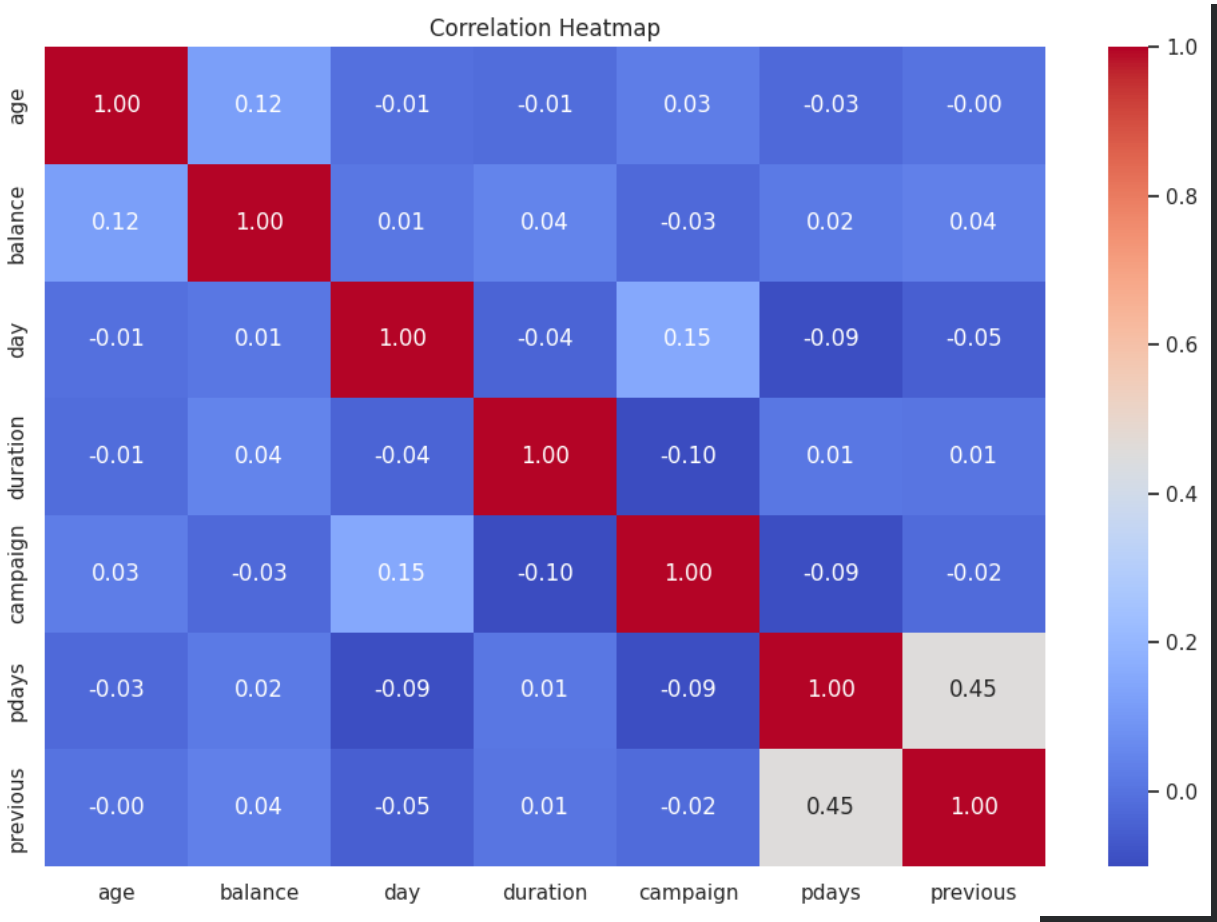
Target distribution:
y
no     39922
yes     5289
Name: count, dtype: int64

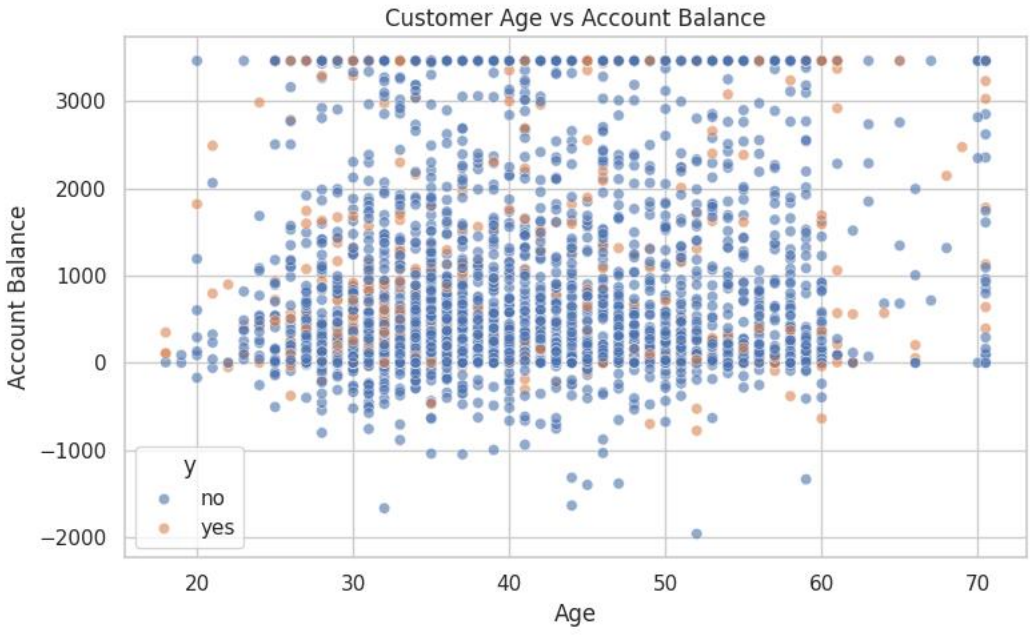
Target percentage:
y
no     88.3
yes    11.7
Name: proportion, dtype: float64
```

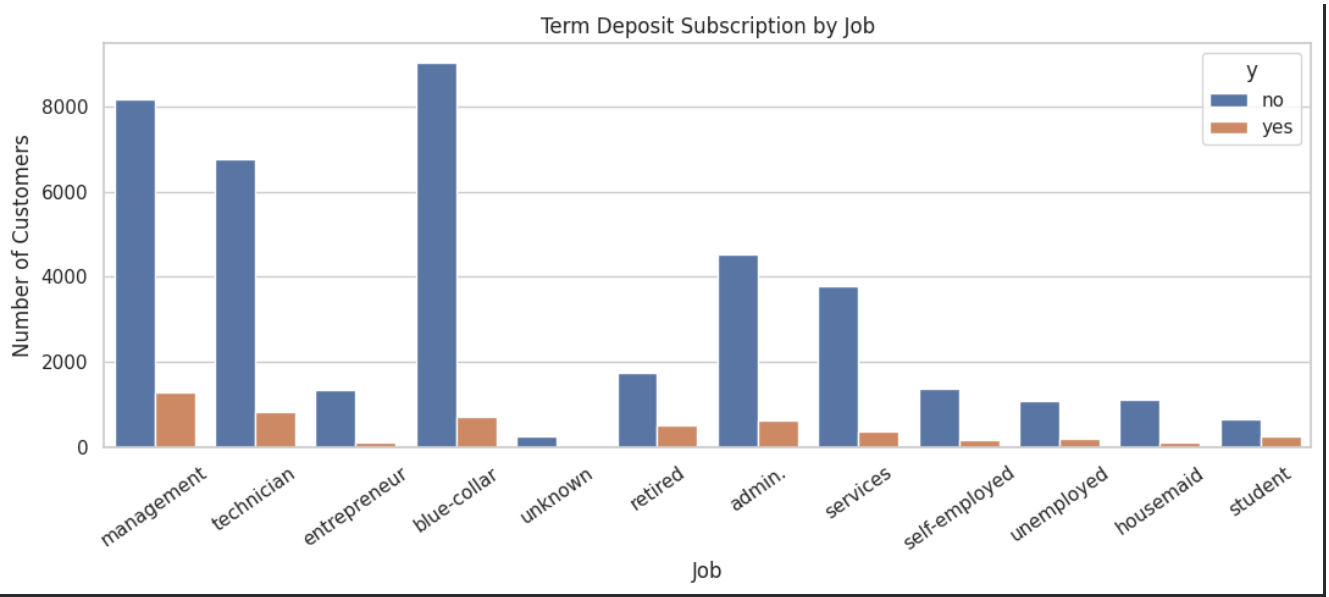



Distribution of Customer Age
Frequency
Age
0
500
1000
1500
2000
2500
3000
3500
4000
20
30
40
50
60
70


Boxplot of Numerical Features
3000
2000
1000
0
−1000
−2000
age
balance
duration
campaign


Correlation Heatmap
age
balance
day
duration
campaign
pdays
previous
1.00 0.12 -0.01 -0.01 0.03 -0.03 -0.00
0.12 1.00 0.01 0.04 -0.03 0.02 0.04
-0.01 0.01 1.00 -0.04 0.15 -0.09 -0.05
-0.01 0.04 -0.04 1.00 -0.10 0.01 0.01
0.03 -0.03 0.15 -0.10 1.00 -0.09 -0.02
-0.03 0.02 -0.09 0.01 -0.09 1.00 0.45
-0.00 0.04 -0.05 0.01 -0.02 0.45 1.00
1.0
0.8
0.6
0.4
0.2
0.0


Customer Age vs Account Balance
Account Balance
3000
2000
1000
0
−1000
−2000
y
no
yes
20
30
40
50
60
70
Age




```
Training samples: 31647
Validation samples: 6782
Testing samples: 6782
Processed feature count: 51

Training model 1 {'units': 32, 'dropout': 0.2, 'learning_rate': 0.001}
Epoch 1/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3s 7ms/step - accuracy: 0.7057 - loss:
0.5463 - val_accuracy: 0.8266 - val_loss: 0.3992
Epoch 2/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.7940 - loss:
0.4281 - val_accuracy: 0.8232 - val_loss: 0.3711
Epoch 3/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8021 - loss:
0.4012 - val_accuracy: 0.8235 - val_loss: 0.3585
Epoch 4/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8061 - loss:
0.3858 - val_accuracy: 0.8251 - val_loss: 0.3517
Epoch 5/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8083 - loss:
0.3789 - val_accuracy: 0.8253 - val_loss: 0.3507
Epoch 6/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8059 - loss:
0.3724 - val_accuracy: 0.8247 - val_loss: 0.3514
Epoch 7/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8127 - loss:
0.3638 - val_accuracy: 0.8309 - val_loss: 0.3387
Epoch 8/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.8090 - loss:
0.3652 - val_accuracy: 0.8324 - val_loss: 0.3385
Epoch 9/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8124 - loss:
0.3624 - val_accuracy: 0.8328 - val_loss: 0.3337
Epoch 10/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step - accuracy: 0.8165 - loss:
0.3568 - val_accuracy: 0.8306 - val_loss: 0.3373
Epoch 11/20
```

```
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8179 - loss:
0.3524 - val_accuracy: 0.8331 - val_loss: 0.3349
Epoch 12/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.8176 - loss:
0.3498 - val_accuracy: 0.8304 - val_loss: 0.3394
Validation AUC: 0.9236

Training model 2 {'units': 64, 'dropout': 0.2, 'learning_rate': 0.001}
Epoch 1/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3s 9ms/step - accuracy: 0.7387 - loss:
0.5022 - val_accuracy: 0.8294 - val_loss: 0.3816
Epoch 2/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.8023 - loss:
0.4023 - val_accuracy: 0.8316 - val_loss: 0.3518
Epoch 3/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8053 - loss:
0.3832 - val_accuracy: 0.8265 - val_loss: 0.3418
Epoch 4/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8088 - loss:
0.3718 - val_accuracy: 0.8260 - val_loss: 0.3425
Epoch 5/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8107 - loss:
0.3648 - val_accuracy: 0.8288 - val_loss: 0.3354
Epoch 6/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8143 - loss:
0.3569 - val_accuracy: 0.8318 - val_loss: 0.3349
Epoch 7/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8170 - loss:
0.3497 - val_accuracy: 0.8318 - val_loss: 0.3348
Epoch 8/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8203 - loss:
0.3428 - val_accuracy: 0.8354 - val_loss: 0.3286
Epoch 9/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8207 - loss:
0.3433 - val_accuracy: 0.8400 - val_loss: 0.3206
Epoch 10/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8247 - loss:
0.3350 - val_accuracy: 0.8390 - val_loss: 0.3232
Epoch 11/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8256 - loss:
0.3321 - val_accuracy: 0.8394 - val_loss: 0.3213
Epoch 12/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8255 - loss:
0.3309 - val_accuracy: 0.8366 - val_loss: 0.3242
Validation AUC: 0.9272

Training model 3 {'units': 64, 'dropout': 0.3, 'learning_rate': 0.0005}
Epoch 1/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2s 6ms/step - accuracy: 0.6630 - loss:
0.6044 - val_accuracy: 0.7955 - val_loss: 0.4456
Epoch 2/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.7741 - loss:
0.4577 - val_accuracy: 0.8133 - val_loss: 0.3872
Epoch 3/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.7863 - loss:
0.4241 - val_accuracy: 0.8130 - val_loss: 0.3795
Epoch 4/20
```

```
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.7958 - loss:
0.4047 - val_accuracy: 0.8201 - val_loss: 0.3606
Epoch 5/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.7996 - loss:
0.3933 - val_accuracy: 0.8191 - val_loss: 0.3623
Epoch 6/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step - accuracy: 0.7997 - loss:
0.3883 - val_accuracy: 0.8204 - val_loss: 0.3562
Epoch 7/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.8022 - loss:
0.3790 - val_accuracy: 0.8197 - val_loss: 0.3523
Epoch 8/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8039 - loss:
0.3773 - val_accuracy: 0.8173 - val_loss: 0.3539
Epoch 9/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8063 - loss:
0.3703 - val_accuracy: 0.8186 - val_loss: 0.3514
Epoch 10/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8097 - loss:
0.3645 - val_accuracy: 0.8229 - val_loss: 0.3439
Epoch 11/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8090 - loss:
0.3630 - val_accuracy: 0.8248 - val_loss: 0.3395
Epoch 12/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8094 - loss:
0.3600 - val_accuracy: 0.8244 - val_loss: 0.3389
Epoch 13/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8126 - loss:
0.3560 - val_accuracy: 0.8232 - val_loss: 0.3431
Epoch 14/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8139 - loss:
0.3540 - val_accuracy: 0.8242 - val_loss: 0.3423
Epoch 15/20
124/124 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 4ms/step - accuracy: 0.8127 - loss:
0.3535 - val_accuracy: 0.8257 - val_loss: 0.3389
Validation AUC: 0.9237

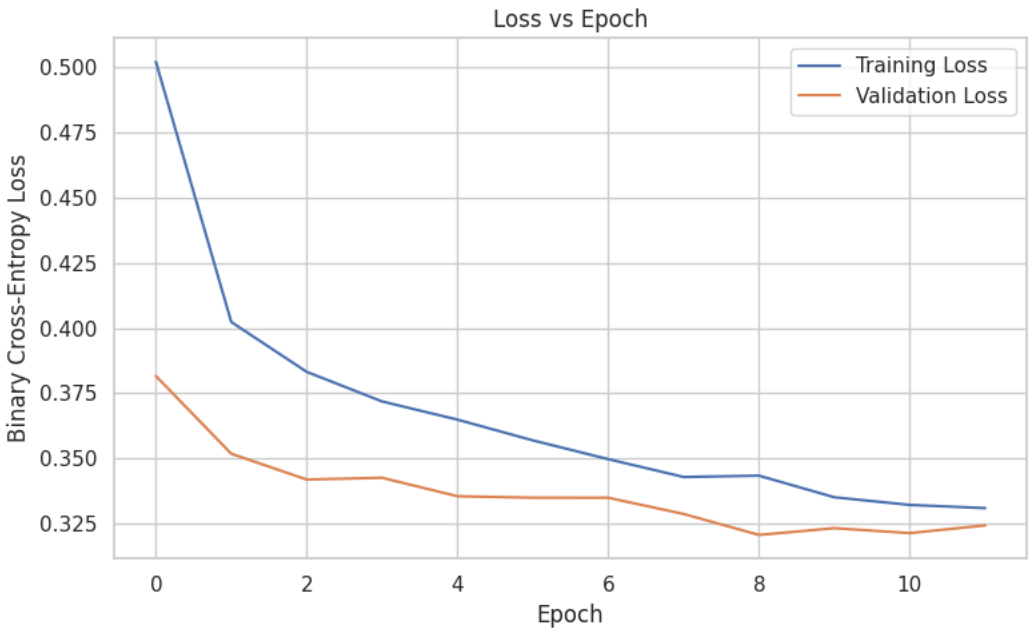
Best settings: {'units': 64, 'dropout': 0.2, 'learning_rate': 0.001}
Best validation AUC: 0.9272

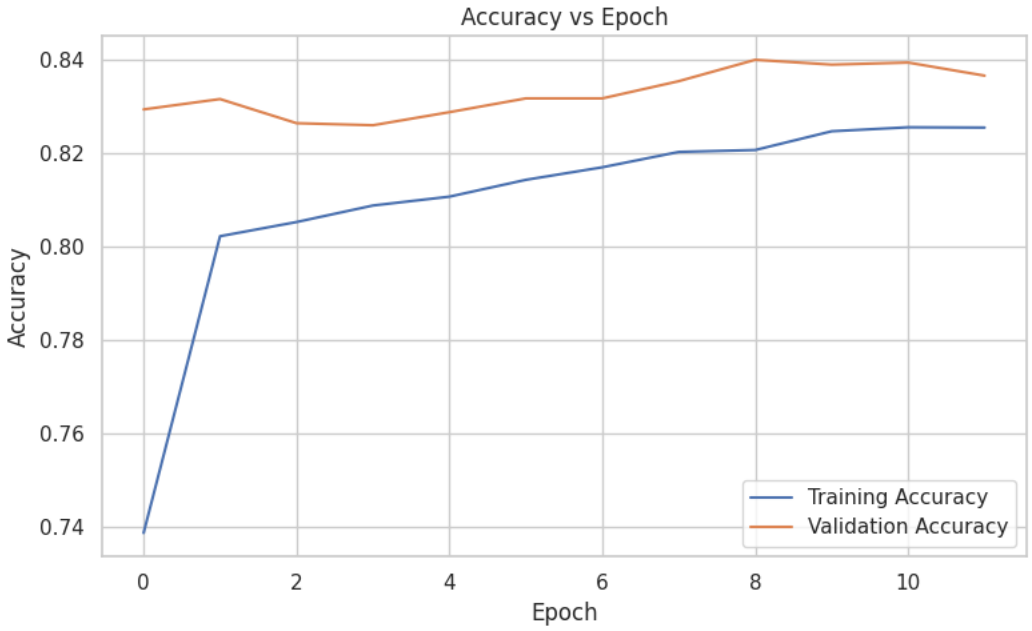
TEST RESULTS
Accuracy: 0.8279
ROC-AUC: 0.9139

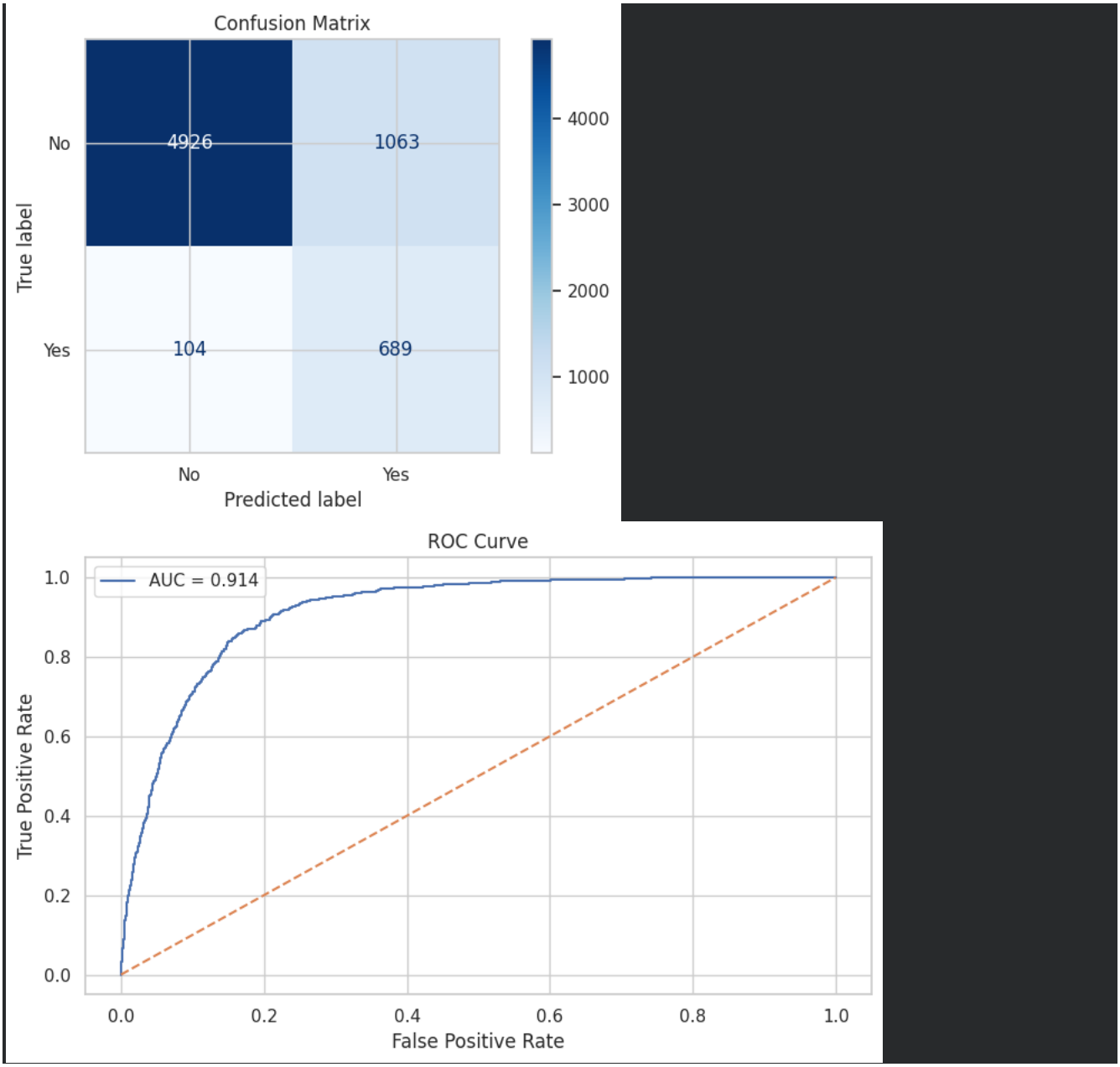
Classification Report:
                precision    recall  f1-score   support

No Subscription       0.98      0.82      0.89      5989
   Subscription       0.39      0.87      0.54       793

       accuracy                           0.83      6782
      macro avg       0.69      0.85      0.72      6782
   weighted avg       0.91      0.83      0.85      6782
```


Loss vs Epoch
Training Loss
Validation Loss
Binary Cross-Entropy Loss
0.500
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0
2
4
6
8
10
Epoch


Accuracy vs Epoch
Training Accuracy
Validation Accuracy
Accuracy
0.84
0.82
0.80
0.78
0.76
0.74
0
2
4
6
8
10
Epoch




```
PROJECT COMPLETED SUCCESSFULLY!
Dataset rows: 45211
Dataset columns: 17
EDA charts: 6
Model evaluation charts: 4
Results saved in bank_marketing_results/
```